Statistics

Chapter 7

Section 7A Revision

Data handling Cycle

1. **Specifying the question**

   What do you want to know?

   - Do adults watch more television than children?
   - Do people remember words better than pictures?

2. **Collecting Data**

   To help answer your questions you need to collect information from:

   - **Primary data**: asking questions yourself by means of a questionnaire or experiment.
   - **Secondary data**: information already collected by someone else in a book or on the internet.

3. **Processing and presenting the data**

   To make sense of your data you need to:

   - **Process the data**: working out percentages, finding frequencies, calculating means, finding the median, mode and range etc.
   - **Present the data**: using frequency chart (bar chart, histogram), pie chart or scatter diagram etc.

4. **Interpreting the data (to answer the question)**

   Finally the results need to be:

   - Examined: for patterns, exceptions, inaccuracies.
   - Interpreted: did you answer your original question.
   - Draw conclusions: give an answer to your question.

Summary of data types

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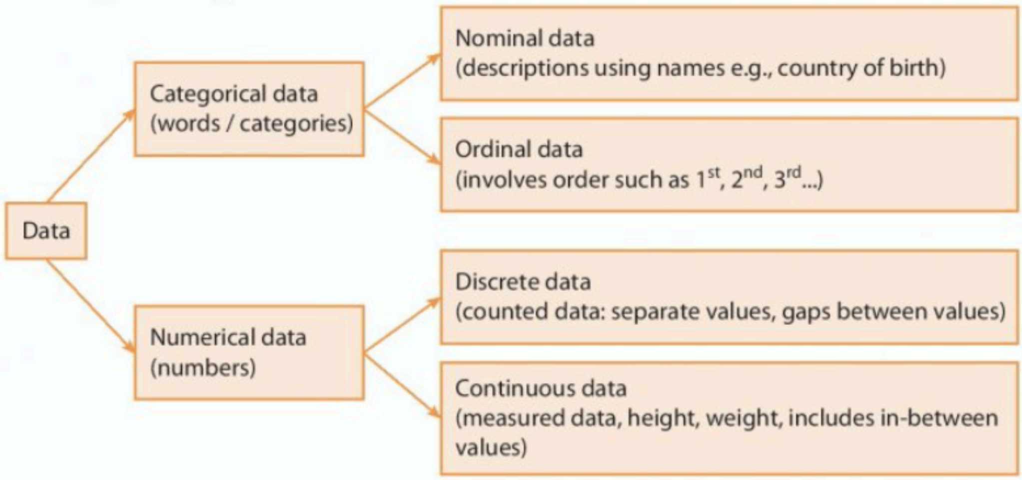

Examples of data

1. **Categorical nominal data:**
   - colour of cars
   - male or female
   - favourite films
2. **Categorical ordinal data:**
   - examination grades
   - opinion scales (strongly agree, agree, don't know...)
   - types of cars (luxury, executive, family, economy...)
3. **Numerical discrete data:**
   - number of children in a family
   - number of desks in a classroom
   - number of goals scored
4. **Numerical continuous data:**
   - heights of students in a class group
   - speeds of cars at a point on the road
   - time taken to complete an exercise

Categorical data can also be described as ***qualitative data*** since it can only be described in words.

Numerical data can be described as ***quantitative data*** since it is defined by numbers.

Exercise 7A

1. In the following list, state whether the data is numerical or categorical:
   (i) the number of points scored in a basketball match
   (ii) the temperature of an oven
   (iii) students' hair colour
   (iv) time taken for athletes to run 400 metres
   (v) star ratings for films
   (vi) countries where cars are manufactured
   (vii) favourite soccer teams of pupils in your class.
   (viii) the number of horses in a race
   (ix) the blood types of students in your class
   (x) the time taken by the winning horse in a race.

2. The following is a list of numerical data. State whether each piece is discrete or continuous:
   (i) Patient's body temperature
   (ii) The score on a die
   (iii) The area of a football pitch
   (iv) The time taken to run a marathon race

3. The following is a list of categorical data. State whether each piece is nominal or ordinal:
   (i) The subjects available to 5th year students
   (ii) The categories of medals that can be won at the Olympic Games
   (iii) The types of animals on a large farm
   (iv) Grades received in Junior Certificate examinations.

4. A (**premier League**) team has (**24**) (**first team**) players.
   Their average age is (**26 years**).
   They were born in (**Ireland, England, Croatia, Senegal, France and the Ivory Coast**).
   Classify each piece of the data in brackets above.

5. (Eight candidates) were interviewed for a job.
Based on the results of the interview, they were placed in order (from 1 to 8).
Describe the data contained in each statement.

Section 7B Collecting Data

A: **Experiments** can be carried out to test products or the performance of equipment.

B: **Observation** can be made of people or processes.

C: **Questionnaires** can be used to ask people questions directly.

In all methods of collecting data it is essential that the survey or experiment is **fair** and avoids **bias**.

When composing a questionnaire you should:

- Make sure the questions are clear and unambiguous
- Ask short, concise questions
- Start with simple questions
- Provide response boxes where possible; Yes( ) No( )
- Avoid leading questions such as: Don't you agree...
- Avoid personal questions involving, names, age, weight etc.

Sampling

A **census** is a survey in which data are collected from every member of a population of a country. In Ireland a census was taken on 24th April 2016 and previous to that on 10th April 2011.

Because every person is included in a census, it is an enormously expensive task.

It is therefore necessary for most surveys to take a **sample** from the population.

It is assumed that the results from the sample, reflects the whole population.

If 30% of the sample say they will vote in a particular way, then we assume that 30% of the voting population will vote the same way if the sample is representative.

The accuracy of the sampling method depends on how **random** the sample is and the quality of the questions asked.

Avoiding bias in sampling methods

(i) Identify your population
Is your question relevant to your particular school, your town or your county?

(ii) Make sure that the sample size is large enough.

(iii) That each member of the sample is chosen randomly.

For example, in a street survey, the time of day and the location may mean that a particular section of the population is being excluded.

Exercise 7B

1. Terry wants to find out how often adults go to the cinema.
   He includes this question on a questionnaire.
   "How many times do you go to the cinema?"
   Not very often ( )   Sometimes ( )   A lot ( )
   (i) Write down two things wrong with this question.
   (ii) Design a better question for his questionnaire to find out how often adults go to the cinema.

2. Eamonn is trying to find out what people think of the *FFG Political Party*. He is trying to decide between these two questions for his questionnaire.
   Which question should he use? Explain your answer.
   (i) How do you rate the *FFG Party's* economic policies?
   Strongly agree ☐   Agree ☐   Don't know ☐   Dissagree ☐
   Strongly Disagree ☐
   (ii) Do you agree that the *FFG Party* has the best economic policies?
   Yes ☐   No ☐

3. Explain what is wrong with each of the following survey questions.
   Suggest a better alternative for each question.
   (i) What age are you?
   (ii) Would you describe yourself as being well-educated?
   (iii) Normal people like animals.
   Do you like animals?
   (iv) Would you agree that top actors are paid too much?
   (v) Have you ever taken illegal drugs?
   (vi) Where exactly do you live?
   (vii) In view of the huge numbers of road accidents outside this school, do you think the speed limit should be reduced?

4. Explain why the following questions are not suitable for a questionnaire.
   In each case write an improved question.
   (i) What do you think of the new Taoiseach?
   (ii) What type of holiday do you like best?
   (iii) Do you agree that you get too much homework?
   (iv) For how long do you social media each day
   (2–3 hours)   (3–4 hours)   (5–6 hours)?
   (v) Do you or your parents often buy DVDs in a store?

5. A school has 820 students and an investigation concerning the school uniform is being conducted.
   40 students from the school are randomly selected to complete the survey on their school uniform. In this situation,
   (i) what is the population size
   (ii) what is the size of the sample?

6. Melanie wants to find out how often people go to the cinema.
   She gives a questionnaire to all the women leaving a cinema.
   Her sample is biased.
   Give **two** possible reasons why.

7. Amy wants to find out how often people play sport.
   She went to a local sports shop and questioned the people she met.
   Explain why this sample is likely to be biased.

8. Dara wants to find out how people get to work each day.
Which of the following is the most appropriate group to question?
A: Every fourth person at a bus stop.
B: A group of people at lunch break.
C: People arriving late for work.

9. Jennifer wants to investigate if people want longer prison sentences for criminals.
Which of the following is the most appropriate group to question?
A: Members of the gardaí.
B: People at a football match.
C: People who have been in prison.
Explain your answer.

10. Amanda wants to choose a sample of 500 adults from the town where she lives.
She considers these methods of choosing her sample:
Method 1: Choose people shopping in the town centre on Saturday mornings.
Method 2: Choose names at random from the electoral register.
Method 3: Choose people living in the streets near her house.
Which method is most likely to produce an unbiased sample?
Give a reason for your answer.

11. Comment on any possible bias in the following situations:
(i) Sixth-year students only are interviewed about changes to the school uniform.
(ii) Motorists stopped in peak-hour are interviewed about traffic problems.
(iii) Real estate agents are interviewed about the prices of houses.
(iv) A 'who will you vote for?' survey at an expensive city restaurant.

Section 7C Averages

An average tries to represent the full set of data values by a single value.

The most commonly used averages are the median, the mode and the mean.

A: The median
When the data is arranged in order from the smallest to the largest; the median is the **middle** number.

2, 5, 6, 7, 10
median

Example 1

The ages of students on a school bus are:
(i) 15 13 16 15 11 16 16 14 15
(ii) 16 14 16 13 18 15

Find the median age.

Placing the ages in order we get:
(i) 11 13 14 15 15 15 16 16 16
(ii) 13 14 15 16 16 18

The median is the 5th number (the middle number) which is 15.

(Because there are two middle values we add them and divide by 2) $\frac{15+16}{2}$ = Median = 15.5

Example 2

The table below shows the age and height of 6 classmates.
(i) What is the median age?
(ii) What is the median height?

	Andre	Lucy	Conor	Phillipe	Donald
Age (years)	16	15	17	16	15
Height (cm)	162	155	164	173	166

(i) Median age: 15, 15, 16, 16, 17, the median (middle) age is 16 years.
(ii) Median height: 155, 162, 164, 166, 173, the median height is 164 cm.

Exercise 7C

1. By arranging the following sets of data in order, find the median for each.
(i) 3, 4, 1, 5, 6, 4, 7, 4, 3, 5, 4
(ii) 1.3, 4.1, 3.5, 2.6, 3.2
(iii) 18, 3, 16, 10, 11, 18, 16, 21, 15
(iv) 5, 2, 6, 4, 8, 6
(v) 45, 54, 46, 51, 47, 44, 39, 50, 48, 48, 47

Remember; if there are **two middle scores,** then we add them together and divide by 2

2. Below is a frequency table of the results of a maths test with marks from 2 to 6.

Mark	2	3	4	5	6
Freq	1	2	1	4	7

By writing the results in increasing order or otherwise, find the median mark.

3. What is the median percentage of Sheila's last 6 French tests:
65%, 55%, 70%, 67%, 59%, 80%

4. (i) Write out 5 numbers so that the median is 9.
(ii) Write down 7 numbers so that the median is 15.
(iii) Write down 4 numbers so that the median is 5.

5. Peter made the following data chart from his water polo team.

	Peter	Frank	Marcus	TJ	Liam
Age (years)	18	14	19	20	17
Height (cm)	172	159	160	178	166
Weight (kg)	55	49	65	71	68

(i) What is the median age?
(ii) What is the median height?
(iii) What is the median weight?

6. A die was rolled 19 times and the numbers recorded as follows:
2, 5, 3, 4, 5, 6, 4, 4, 2, 1, 3, 5, 3, 2, 5, 1, 5, 6, 3.
(i) Find the median number of the above data.
(ii) If a die is rolled a large number of times what median number would you expect?

7. In Tanya's experiment the median number of peas in a pod was 11.
Find the largest possible value of x.

No. of peas in a pod	10	11	12
No. of pods	8	9	x